

## Formulaire de trigonométrie

## Identités fondamentales

$$\begin{aligned}\cos^2(x) + \sin^2(x) &= 1 \\ 1 + \tan^2(x) &= \frac{1}{\cos^2(x)} \\ 1 + \cot^2(x) &= \frac{1}{\sin^2(x)}\end{aligned}$$

## Formules d'addition

$$\begin{aligned}\cos(a+b) &= \cos(a)\cos(b) - \sin(a)\sin(b) \\ \cos(a-b) &= \cos(a)\cos(b) + \sin(a)\sin(b) \\ \sin(a+b) &= \sin(a)\cos(b) + \cos(a)\sin(b) \\ \sin(a-b) &= \sin(a)\cos(b) - \cos(a)\sin(b) \\ \tan(a+b) &= \frac{\tan(a)+\tan(b)}{1-\tan(a)\tan(b)} \\ \tan(a-b) &= \frac{\tan(a)-\tan(b)}{1+\tan(a)\tan(b)}\end{aligned}$$

## Formules de duplication

$$\begin{aligned}\sin(2x) &= 2\sin(x)\cos(x) \\ \cos(2x) &= \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 \\ \tan(2x) &= \frac{2\tan(x)}{1-\tan^2(x)}\end{aligned}$$

## Formules d'angle moitié

$$\begin{aligned}\sin^2\left(\frac{x}{2}\right) &= \frac{1-\cos(x)}{2} \\ \cos^2\left(\frac{x}{2}\right) &= \frac{1+\cos(x)}{2} \\ \tan\left(\frac{x}{2}\right) &= \frac{\sin(x)}{1+\cos(x)} = \frac{1-\cos(x)}{\sin(x)}\end{aligned}$$

## Produit en somme

$$\begin{aligned}\cos(a)\cos(b) &= \frac{1}{2}[\cos(a-b) + \cos(a+b)] \\ \sin(a)\sin(b) &= \frac{1}{2}[\cos(a-b) - \cos(a+b)] \\ \sin(a)\cos(b) &= \frac{1}{2}[\sin(a+b) + \sin(a-b)]\end{aligned}$$

## Somme en produit

$$\begin{aligned}\cos(a) + \cos(b) &= 2\cos\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right) \\ \cos(a) - \cos(b) &= -2\sin\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right) \\ \sin(a) + \sin(b) &= 2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right) \\ \sin(a) - \sin(b) &= 2\cos\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right)\end{aligned}$$

## Formules liées aux angles remarquables

$$\begin{aligned}\cos(\pi - x) &= -\cos(x) \\ \sin(\pi - x) &= \sin(x) \\ \tan(\pi - x) &= -\tan(x) \\ \cos(\pi + x) &= -\cos(x) \\ \sin(\pi + x) &= -\sin(x) \\ \tan(\pi + x) &= \tan(x) \\ \cos\left(\frac{\pi}{2} - x\right) &= \sin(x) \\ \sin\left(\frac{\pi}{2} - x\right) &= \cos(x) \\ \tan\left(\frac{\pi}{2} - x\right) &= \cotan(x) \\ \cos\left(\frac{\pi}{2} + x\right) &= -\sin(x) \\ \sin\left(\frac{\pi}{2} + x\right) &= \cos(x) \\ \tan\left(\frac{\pi}{2} + x\right) &= -\cotan(x)\end{aligned}$$